PNAS RESEARCH ARTICLE | BIOLOGICAL SCIENCES

Machine-learning dissection of endogenous A-to-I RNA editing reveals substrate- and target-level determinants of predictability

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This manuscript was compiled on August 25, 2026

Endogenous adenosine-to-inosine RNA editing reflects at least three scales: the intrinsic competence of an RNA duplex, its realization and encounter with ADAR in a cell, and selection of an individual adenosine. Which of these scales contains recoverable information beyond current sequence–structure models is unknown. We used machine learning as a controlled instrument for information dissection, evaluating biological annotations, supervision, representation, transfer, capacity, and integration on 884 human inverted-Alu substrates while quarantining the published pair-disjoint test. Substrate-level editing burden was predictable, and inter-arm geometry added reproducible information beyond graph-derived summaries, consistent with variation in realization or engagement. The tested RBP, transcript, repeat, and bulk chromatin measurements did not resolve that underlying cellular state sufficiently to improve site prediction. Two actionable principles emerged. Sampling substrates before sites prevented site-rich duplexes from dominating learning, separately from the benefit of recovering additional high- and low-editing observations. A target/helix-partner-aligned representation recovered local selectivity diluted by whole-duplex pooling; an identical encoder shifted 96 nucleotides away failed. A fixed equal-logit combination of independently trained whole-duplex-only and local-augmented graph predictors outperformed a preregistered unified network in development. Frozen before test access, ADARPro improved AUROC, AUPRC, and F1 in all five reported benchmark endpoints; mean improvements were 0.0277, 0.0261, and 0.0253, with positive pair-clustered intervals throughout. Endogenous editing predictability is hierarchical: substrate-level propensity is detectable but incompletely resolved by bulk measurements, whereas substrate-aware learning and target-centered representation provide actionable information.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Significance

Why one apparently suitable adenosine is edited while another is not remains unclear. Using a pair-disjoint machine-learning framework, we separated information carried by the RNA duplex, substrate-level editing propensity, and the local environment of the target nucleotide. Editing burden showed a reproducible component associated with pair geometry, yet current tissue-average annotations did not resolve the underlying cellular state. Learning equally from RNA substrates and explicitly aligning each target with its helix partner produced the strongest gains. The study identifies both what present data can predict and the matched cellular measurements needed next.

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA), diversifying transcripts while preventing endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Nevertheless, editing is selective: nearby adenosines within the same predicted structure can differ sharply, and the same locus can vary across tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context contribute to this selectivity (13–17), but they do not specify whether a potential substrate forms, persists, localizes appropriately, or encounters the relevant ADAR state.

Recent structure-aware predictors, exemplified by AdarEdit, model intrinsic editability over complete inverted-Alu substrates (18). Its compact graph attention network encodes nucleotide identity, local and global sequence, predicted secondary structure, pairing topology, geometry, thermodynamics, and motifs. A strict benchmark assigns entire Alu pairs globally to train, validation, or test, avoiding the severe leakage that would occur if sites from one duplex crossed partitions. Residual errors could therefore reflect missing

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Competing-interest declaration pending author confirmation.

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cellular context, underused supervision, pooling that dilutes the focal nucleotide, or an inadequate separation of substrate-level propensity from local selectivity.

These explanations cannot be distinguished by adding one more annotation or a larger network. A biologically relevant variable can be redundant with sequence, measured at the wrong cellular scale, or too noisy to improve prediction. An architectural gain can likewise arise from overweighting substrates with many labeled sites rather than from a better account of editing. We therefore asked a broader question: within measurements available to the benchmark, where does incremental predictive information for endogenous A-to-I editing reside?

We preserved the published benchmark and examined biological context, supervision, substrate-wide and target-local representations, cross-tissue transfer, RNA foundation models, hierarchical formulations, and two forms of integration. Machine learning served both as predictor and controlled assay: changing supervision, representation, transfer, or integration tested which biological scales contain recoverable information beyond the sequence–structure graph. Every discovery comparison used pair-disjoint development folds and pair-clustered uncertainty; the canonical test remained sealed until one model was frozen. The evidence resolved four principles. Whole-duplex sequence and structure encode intrinsic competence. Substrate-level editing burden contains a reproducible component consistent with realization or engagement, although current bulk context proxies do not resolve the underlying cellular state. A target and its helix partner retain local information that whole-duplex pooling loses. Finally, the duplex is the effective statistical unit for learning from correlated sites. ADARPro follows this hierarchy rather than increasing capacity indiscriminately.

Results

A controlled framework separates biological scales. The benchmark comprises four individual GTEx tissue endpoints—artery tibial, brain/cerebellum, liver, and skeletal muscle—and Combined, a pooled endpoint rather than a fifth biological tissue. EditingIndex values at least 15% define positives and values below 1% define negatives; intermediate observations are excluded from the canonical binary task. The fixed global split contains 566 training, 139 validation, and 179 test pairs with zero overlap (Fig. 1).

Within the training and validation pairs, 5 outer folds generated out-of-fold (OOF) predictions for hypothesis testing and model selection. Feature fitting, checkpoint selection, and threshold choice were isolated from each outer fold. The internal reference was always trained and evaluated under the same folds as its candidate. Its lower absolute score reflects reduced training data and is not a replacement for published AdarEdit performance. The supplied-checkpoint canonical test remained the authoritative benchmark and was accessed only after the final model, thresholds, weights, code, and hashes had been frozen.

The campaign mapped hypotheses onto a biological hierarchy (Fig. 1A). Intrinsic duplex properties and cell- or transcript-specific state jointly govern whether a substrate is available to editing; local sequence and pairing then help distinguish adenosines within it. Outcomes were classified as selected, supported but not yet actionable, limited, not supported in the tested formulation, deferred for lack of matched measurements, or not pursued after stronger lower-complexity evidence emerged (Fig. 2A). These categories separate biological relevance from incremental utility at the resolution tested here.

The substrate is a biological and statistical unit. The canonical row set samples a subset of the available observations, whereas the underlying training tables contain additional sites with valid high or low EditingIndex. Recovering these labels improves supervision, but uniform site sampling gives a substrate weight proportional to its number of observations. Sites within one duplex share sequence, structure, genomic context, and often editing behavior; they are not independent biological units.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 3A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). The two corrections therefore make separable contributions.

The combined pair-first formulation improved the same-fold reference in all five endpoints (Fig. 3B). All 15 AUROC, AUPRC, and F1 intervals excluded zero, with F1 gains from +0.69 to +4.23 percentage points. This is a learning result rather than a claim that the pair is a single causal molecular determinant: pair-first sampling corrects the effective contribution of correlated sites.

The substrate also carries biological variation in editing burden. Genomic separation between Alu arms—information discarded when model inputs join them with a fixed separator—correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 2B). Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 2C; SI Appendix). Pair geometry thus identifies a reproducible substrate-level component of editing burden, consistent with variation in realization or engagement but not a direct measurement of either process.

Bulk context does not resolve the latent cellular state. Making substrate-level propensity actionable for site prediction proved harder than detecting it. A cross-fitted engagement gate added 0.0084 AUPRC but only 0.0041 F1, whose interval included zero, and increased Brier score by 0.0051. A neural engagement auxiliary likewise failed its prespecified promotion criterion. Static geometry marks a stable propensity; it cannot reveal whether a particular transcript formed the expected duplex, remained accessible, occupied the right compartment, or encountered active ADAR.

The initial cellular-context hypothesis focused on RBP competition. We combined replicated ENCODE eCLIP occupancy, PrismNet and RBPNet predictions, literature-prioritized regulators, GTEx tissue abundance, binding-by-abundance interactions, and randomized or shuffled controls (19–22). Alu multimapping makes a

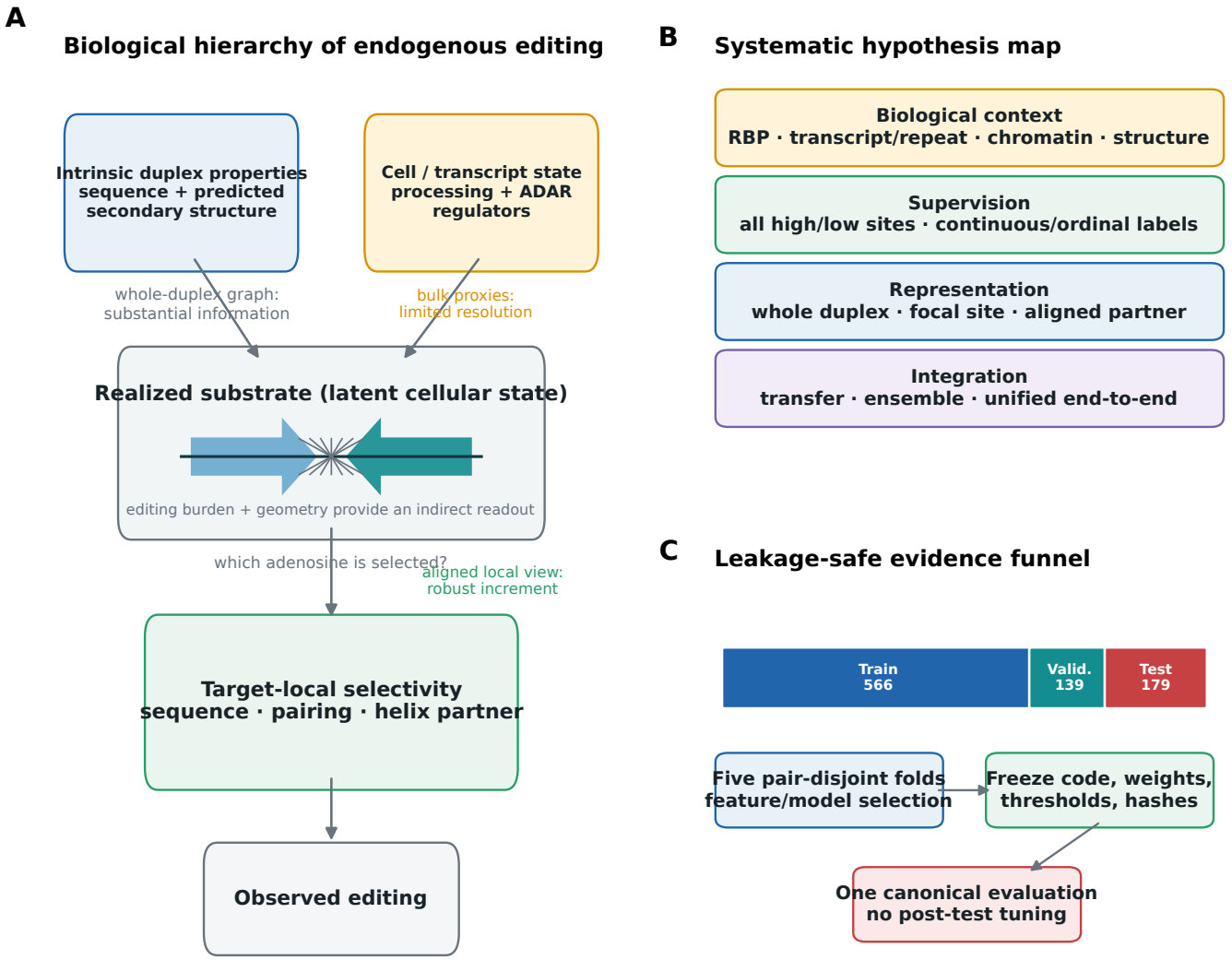


Fig. 1. A biological hierarchy and leakage-safe dissection of editing predictability. (A) Intrinsic duplex properties and cell/transcript state converge on whether an inverted-Alu substrate is available to editing; target/helix-partner context then contributes to selection of a particular adenosine. Pair-level editing burden provides an indirect readout of substrate propensity, whereas the underlying cellular realization state was not directly measured. (B) Controlled AI interventions tested biological context, supervision, representation, and integration. (C) Whole pairs were partitioned globally. All discovery used pair-disjoint development folds; the final specification was frozen before one canonical held-out evaluation. Combined was never assigned a fabricated tissue state.

missing eCLIP peak ambiguous; sequence-based predictors supplied a complementary test independent of peak absence. Neither cell-line occupancy nor tissue-average abundance contributed reproducible residual information. Cross-tissue label switches were also nearly separated by tissue identity alone, preventing four tissue means from identifying an RBP-specific transferable mechanism. These measurements are therefore unlikely to resolve transcript-specific regulator exposure; matched cell- or donor-level assays are the more informative next test (23–26).

Broad transcript, repeat-subfamily, divergence, and neighborhood annotations likewise failed to improve site-level prediction beyond the graph. Exact-tissue bulk H3K36me3 overlap produced a small liver AUPRC increment without stable AUROC, F1, or pair-level benefit; no adequately matched H2A.Z resource supported a powered test. The weak bulk H3K36me3 effect points toward quantitative, sample-matched assays of cotranscriptional state rather than binary tissue-level overlap (27, 28). Together, these results sharpen the measurement problem: substrate-level propensity is detectable, but available static and tissue-average proxies do not resolve the cellular state needed to improve the strongest site classifier.

Target alignment resolves site-level selectivity. The compact AdarEdit graph provides a strong account of whole-duplex competence, but averaging nucleotide embeddings can dilute the state of one target. Two intermediate experiments localized the missing information. Simple target-contact geometry improved ranking and calibration but not F1, and explicitly retaining the focal graph node improved AUROC and AUPRC more consistently than an otherwise matched duplicate-global control. Both results implicated target position before a specialized local encoder was built.

The selected local-augmented graph predictor adds a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nu-

A

Evidence map: what the benchmark can and cannot currently resolve

Scale	Tested formulation	Status	Interpretation
Substrate-aware learning	Pair-first expanded supervision	Selected	Corrects label use and pair weighting
Target selectivity	Aligned target + helix-partner view	Selected	Survives shifted-window control
Integration	Fixed whole-duplex + local-augmented logits	Selected	Complementary same-fold predictions
Substrate propensity	Pair geometry / editing burden	Supported*	Editing burden is learnable; latent state unresolved
Supervision	Continuous + ordinal auxiliary labels	Limited	Ranking benefit; no useful F1 gain
Target selectivity	Focal node / contact geometry	Limited	Points toward aligned local view
Cross-tissue	Shared four-tissue training	Limited	Transfer below promotion criterion
Substrate propensity	Hierarchical engagement gate	Limited	Small ranking gain; unstable classification
Biological context	Bulk H3K36me3 overlap	Limited	Context-specific AUPRC only
Biological context	RBP binding × bulk expression	Not supported	No reproducible residual gain
Biological context	Transcript / repeat annotation panel	Not supported	No robust site-level increment
Representation	Frozen RNA-language embeddings	Not supported	No consistent incremental utility
Representation	Substrate-wide all-A decoder	Not supported	Stopped at prespecified futility gate
Capacity	Larger / deeper graph networks	Not supported	Capacity alone did not generalize
Integration	Unified end-to-end architecture	Not supported	Inferior to frozen ensemble
Cellular state	Matched donor/cell multimodality	Deferred	Required measurements unavailable
RNA state	Matched in vivo structure / isoforms	Deferred	Required measurements unavailable
Capacity	Graph Transformer / GATv2 expansion	Not pursued	Superseded by controlled lower-capacity evidence

Status reflects prespecified same-fold comparisons.
*Reproducible observation not yet actionable for site prediction; deferred hypotheses were not tested.

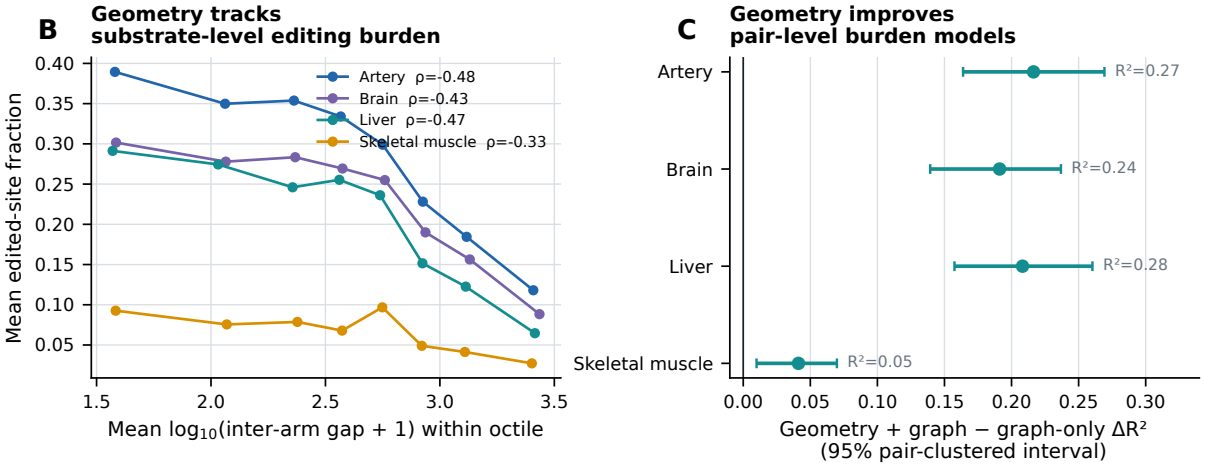


Fig. 2. A structured evidence map and positive evidence for substrate-level propensity. (A) Hypotheses are grouped by biological or learning scale and epistemic outcome. Status reflects prespecified same-fold comparisons and controls; unsupported refers only to the tested formulation, deferred measurements were unavailable, and superseded branches were not interpreted as negative. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 from adding genomic geometry to graph-derived summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

cleotide identity, pairing state, focal and partner indicators, and signed relative position; its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data it reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 3C).

The decisive control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. The matched-capacity shift collapses performance, localizing the incremental information to the target-centered representation rather than to added convolutional capacity. Replication in liver and secondary natural-prevalence analysis supported the same direction.

Whole-duplex and target-local information are complementary. The local-augmented predictor did not make an independently optimized whole-duplex view redundant. Their errors remained sufficiently distinct that a fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver; all six pair-clustered intervals excluded zero (Fig. 3D).

ADARPro was consequently specified as two independently trained pair-first components: a whole-duplex-only graph predictor and a second graph predictor internally augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed, unsearched 0.5/0.5 mean (Fig. 3E). This design tests whether independently optimized global and target-local information remain complementary without fitting a flexible stacker to only hundreds of independent substrates.

Additional complexity provides limited returns. Intermediate EditingIndex values contain graded information, but most useful thresholded content was already recovered by pair-first binary training. A binary/ordinal/continuous multitask head increased AUROC by 0.0081 (95% pair-clustered interval 0.0010–0.0151) while changing F1 by only 0.0008; its integration was adverse under natural prevalence. Shared four-tissue training likewise showed transferable rules, with mean changes across the four individual tissue endpoints of 0.0087 AUROC, 0.0114 AUPRC, and 0.0075 F1, but missed its prespecified promotion criteria and worsened mean Brier score by 0.0043. Graded labels and cross-tissue sharing therefore refine ranking, but less than correcting supervision and target representation.

Other increases in capacity were unproductive. A substrate-wide decoder trained all candidate adenosines jointly changed AUROC by -0.0664 and F1 by -0.0599 relative to the pair-first model and met its prespecified two-fold futility boundary. Frozen RiNALMo target and substrate embeddings, including a larger model, added little beyond the task-specific structure graph (29). Larger or deeper graph networks likewise failed stronger-model controls. More parameters were not a substitute for the correct information structure. The more pertinent pretraining question—benchmark-safe, inverted-Alu-specific representation learning—remained deferred because a leakage-safe corpus could not be guaranteed.

Before test access, we preregistered one connected architecture to ask whether the surviving principles interact more effectively during joint optimization. It combined a three-layer graph encoder, focal-node state, aligned local encoder, pairwise representation interactions, and a training-only substrate head. Its 65,698 trainable parameters were fewer than the 80,138 parameters across the two ensemble components. On the 5,967-row, 585-pair muscle development set, the unified model reached 0.9072/0.8922/0.8317 AUROC/AUPRC/F1 versus 0.9142/0.9040/0.8347 for the frozen ensemble. AUROC and AUPRC were worse in every fold; their pair-clustered intervals were wholly adverse, and Brier increased by 0.0205 (0.0135–0.0276). Prespecified repair and replication branches remained closed. The empirical conclusion is specific: the tested joint architecture did not outperform fixed prediction-level integration at this effective sample size.

A predictor built from the hierarchy improves every canonical endpoint. After development selection, 10 component fits were trained on published training pairs. Published validation data alone selected checkpoints, epochs, thresholds, and one threshold per endpoint. The exact code revision, configurations, weights, and hashes were bound into a final manifest. A single authorization then exposed the canonical test for one evaluation; no model or claim-defining analysis was changed afterward.

ADARPro improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 4A). Mean gains across those endpoints over reproduced supplied-checkpoint AdarEdit were 0.0277, 0.0261, and 0.0253. Because Combined overlaps the individual tissue endpoints, this mean is descriptive rather than a five-tissue meta-analysis. F1 changed from 0.8671 to 0.8869 in artery, 0.8689 to 0.8902 in brain, 0.8541 to 0.8812 in liver, 0.8139 to 0.8541 in skeletal muscle, and 0.8674 to 0.8852 in Combined. Every endpoint had a positive 95% pair-clustered bootstrap interval for all three metrics, and Brier score decreased throughout (Fig. 4C).

The gains were not purchased by indiscriminate suppression of positive calls. At frozen validation-selected thresholds, ADARPro corrected 32.0–50.6% of baseline false positives while damaging 1.7–5.9% of baseline true positives (Fig. 4D). In a post-test descriptive analysis that rethresholded ADARPro to match AdarEdit recall exactly, the point estimates were 126, 119, 158, 68, and 148 fewer false positives across artery, brain, liver, skeletal muscle, and Combined; every pair-clustered interval excluded zero (Fig. 4E). The canonical gain provides an out-of-sample test of the predictive consequences of the hierarchy inferred entirely in development.

Discussion

This hierarchy describes recoverable predictive information rather than a complete causal decomposition of RNA-editing biology. AdarEdit already captures substantial intrinsic duplex competence. The same benchmark nevertheless contains an additional substrate-level dimension: inter-arm geometry covaries with editing burden, and low-capacity pair models predict that burden beyond graph-only summaries. Conditional on a plausible substrate, the fate of an adenosine retains target/helix-partner information that global pooling dilutes. Sites sharing a duplex also form a correlated statistical unit; learning improves when substrates, rather than abundant rows, control effective weight.

These levels should not be conflated. Pair-first sampling is a statistical correction motivated by biological dependence, not evidence that all sites in a substrate share one causal fate. Conversely, the pair-level findings are positive biological observations even though their integration did not improve the leading site classifier. Geometry estimates a stable propensity. Actual realization and engagement also depend on transcription, splicing, residence time, localization, in vivo conformation, ADAR isoform and activity, and regulator occupancy—states averaged away by tissue medians and binary peaks.

The limited and unsupported branches locate the remaining measurement gap. Cell-line RBP occupancy combined with tissue-median abundance supplied no reproducible residual information about transcript-specific regulator exposure. The weak bulk H3K36me3 effect makes quantitative, sample-matched cotranscriptional measurements the sharper next test. Cross-tissue sharing and continuous labels yielded small ranking effects, showing that transferable and graded structure exists but contributes less than corrected supervision and target alignment. Generic frozen RNA embeddings and greater

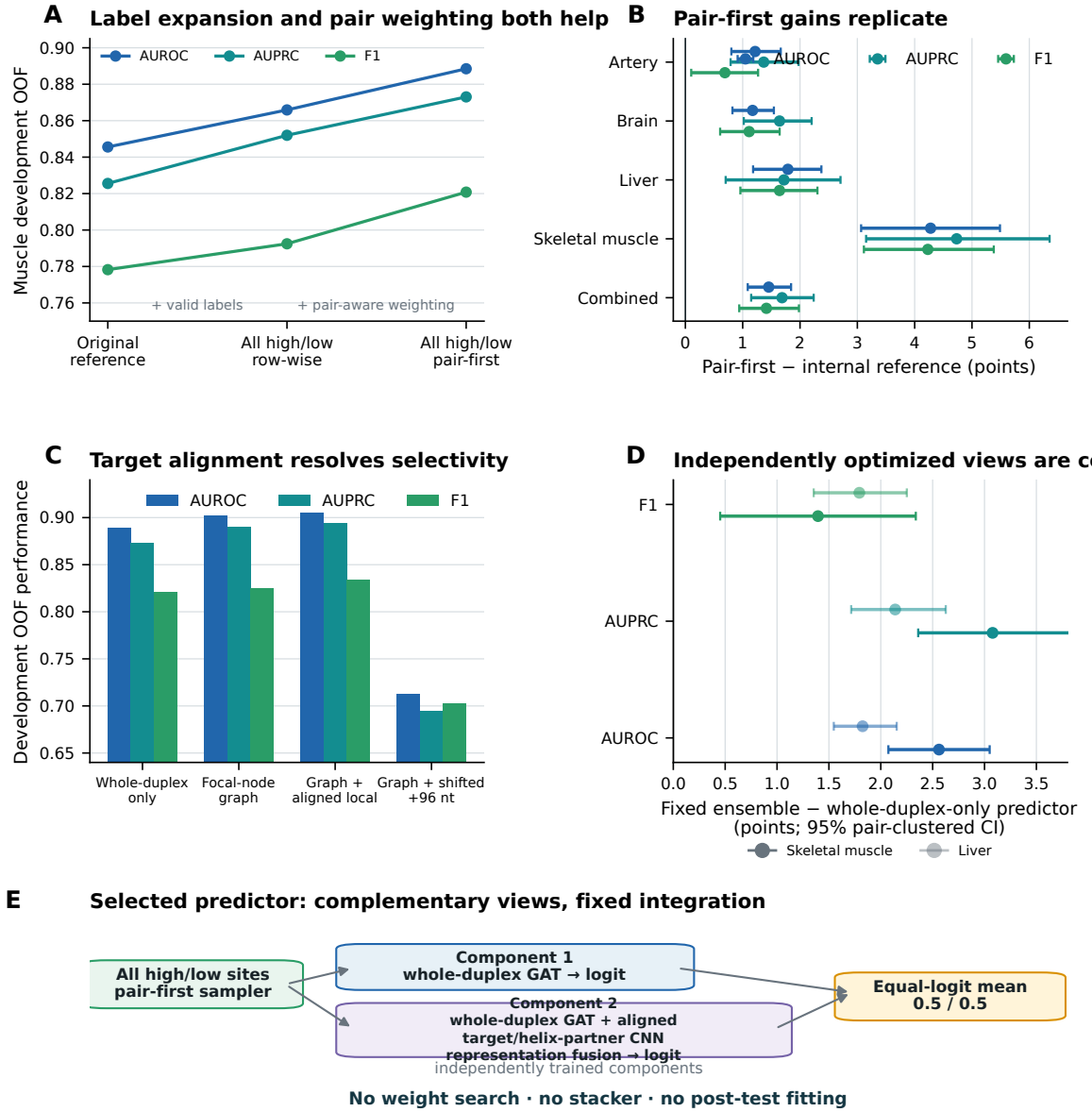


Fig. 3. The two actionable AI principles and their integration. (A) Same-fold ablations separate the benefit of recovering additional valid high/low labels under row-wise sampling from the further benefit of pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, explicit focal-state retention, and graph-plus-aligned-local fusion form an evidence progression; an otherwise identical local-augmented model shifted by 96 nt fails. (D) The fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) Exact selected predictor: an independently trained whole-duplex-only graph model and graph-plus-local fusion model, followed by a fixed 0.5/0.5 logit mean. No ensemble weight, stacker, or post-test parameter is fitted. All panels show controlled development evidence.

graph capacity likewise did not substitute for information localized to the biological decision.

The preregistered unified experiment qualifies another common assumption: representation-level integration is not automatically superior to fixed prediction-level integration. The tested connected architecture ranked and calibrated less well than the frozen ensemble, despite incorporating the validated views and using fewer total parameters. The comparison does not identify the optimization mechanism. It does establish why the final predictor is deliberately simple: a whole-duplex-only graph and an independently trained graph-plus-local model, combined without a learned weight.

Several limitations define the scope of the conclusions. The benchmark contains graph-buildable human inverted-Alu substrates and tissue summaries, not donor-resolved measurements or all editing substrates. Predicted secondary structure is a surrogate for the conformations available to ADAR in vivo. The canonical task excludes intermediate observations and is balanced; natural-prevalence analyses are secondary. eCLIP coverage is cell-line-specific and compromised by repeat mapping. Finally, the canonical partition is stringent but derives from the same benchmark construction; validation in an independent cohort remains necessary.

The canonical gain provides an out-of-sample test of the predictive consequences of this hierarchy. Better

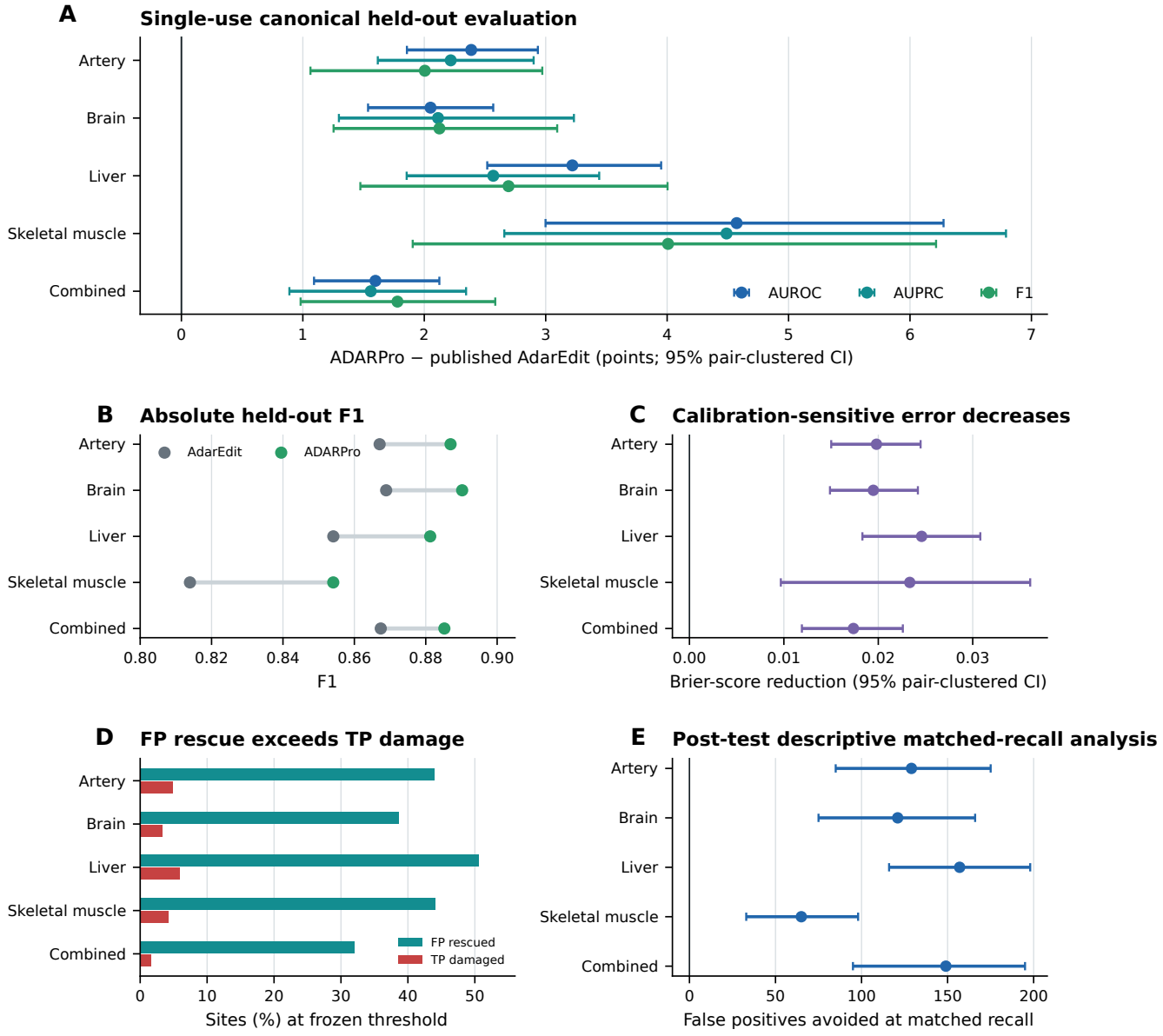


Fig. 4. Single-use canonical held-out evaluation. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score, a calibration-sensitive error measure. (D) Baseline false positives corrected and baseline true positives damaged at frozen validation-selected thresholds. (E) Descriptive post-test false-positive reduction after rethresholding ADARPro to match AdarEdit recall. Combined is a pooled endpoint. All quantities were generated after the final manifest was frozen and the one-use authorization was consumed.

prediction followed from respecting substrate dependence and preserving target-centered context, not from indiscriminate feature or capacity growth. The major unresolved layer is the cellular state underlying substrate-level propensity. The next decisive study should measure editing, transcript abundance and processing, splicing and intron retention, ADAR1 isoform and interferon state, ADAR activity and localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation in the same donors or cells and on the same substrates. Such data could directly test a two-stage model in which cellular state governs whether an intrinsically competent substrate is realized or engaged, and local RNA context governs which adenosine is selected.

Materials and Methods

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human labels were edited for EditingIndex $\geq 15\%$, nonedited for EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. Pair identities and the fixed 566/139/179 train/validation/test assignment were verified before analysis. Combined was treated only as its published pooled endpoint.

Development protocol. All discovery used the original train plus validation pairs divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice were fitted without the outer fold. Predictions were concatenated only after every row was scored by a model that had not trained on its pair. The test paths were protected by a stateful quarantine. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The first component retained the compact AdarEdit graph attention encoder and trained from the expanded high/low site pool using uniform pair-first and equal-class sampling. The second, independently trained component retained the same graph encoder and added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers; local and graph embeddings were fused before its prediction head. The selected probability was $\sigma[(\operatorname{logit} p_{\mathrm{whole}} + \operatorname{logit} p_{\mathrm{augmented}})/2]$; weights were fixed at 0.5/0.5 rather than searched. The shifted-window control preserved the graph-plus-convolution architecture and optimization while moving the local center 96 nt. Full feature definitions, architectures, ablations, and biological-context protocols are in SI Appendix.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each frozen training run. Uncertainty used a pair-clustered bootstrap with 2,000 replicates that resampled Alu pairs and retained all their sites. For the post-test descriptive matched-recall analysis, the candidate threshold was selected on final predictions solely to equal supplied-checkpoint baseline recall; uncertainty again resampled pairs. Before final evaluation, code, inputs, weights, epochs, thresholds, and checkpoint SHA-256 values were frozen. Test authorization was consumed once, and no post-test model selection or tuning occurred. Natural-prevalence results use all eligible high/low observations from development pairs under unchanged pair assignments and are secondary to the canonical balanced task.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

ACKNOWLEDGMENTS. Acknowledgments and funding information pending author confirmation.

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